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Problem Set 3

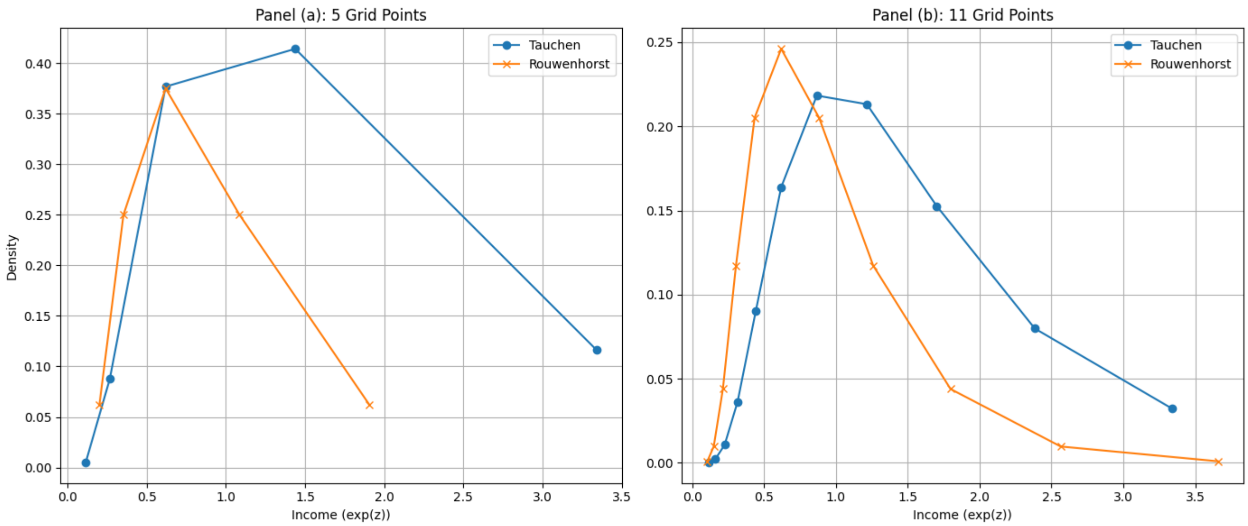
Macroeconomic Theory II
April 24, 2025

Question 1 - Income Process Approximation

The full code for this problem set can be found at the following Google Colab code (link)

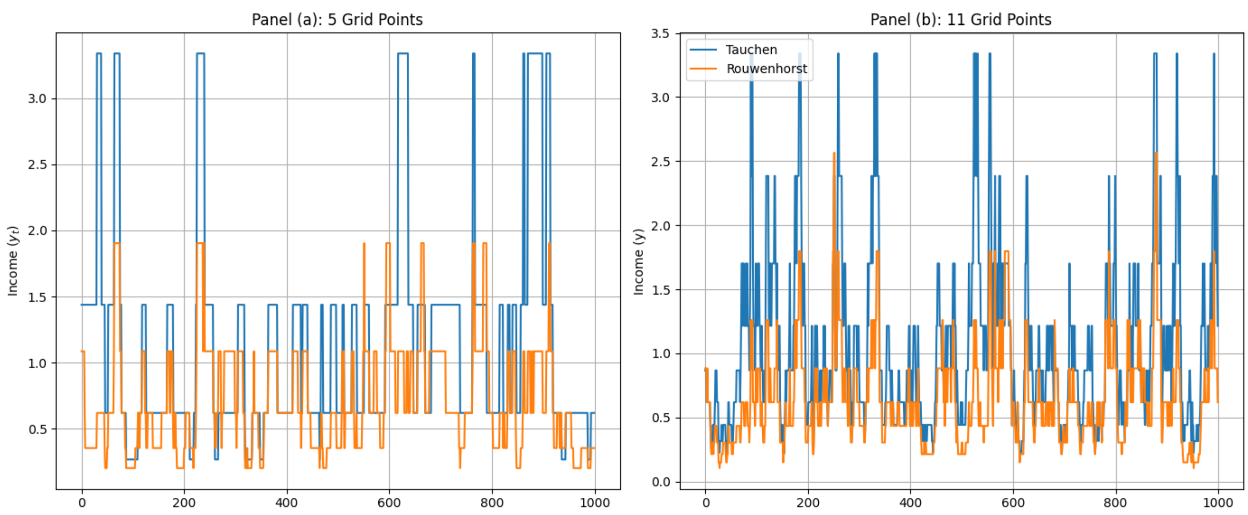
1.1, 1.2 & 1.3- Tauchen Discretization Method

Figure 1: Stationary Distribution: Rouwhenhorst $\times$ Tauchen - $\rho = 0.9$



Note: I Compute the stationary distribution using the eigenvalue method, $\pi'(I - P) = 0$

Figure 2: Income Simulation: Rouwhenhorst $\times$ Tauchen - $\rho = 0.9$

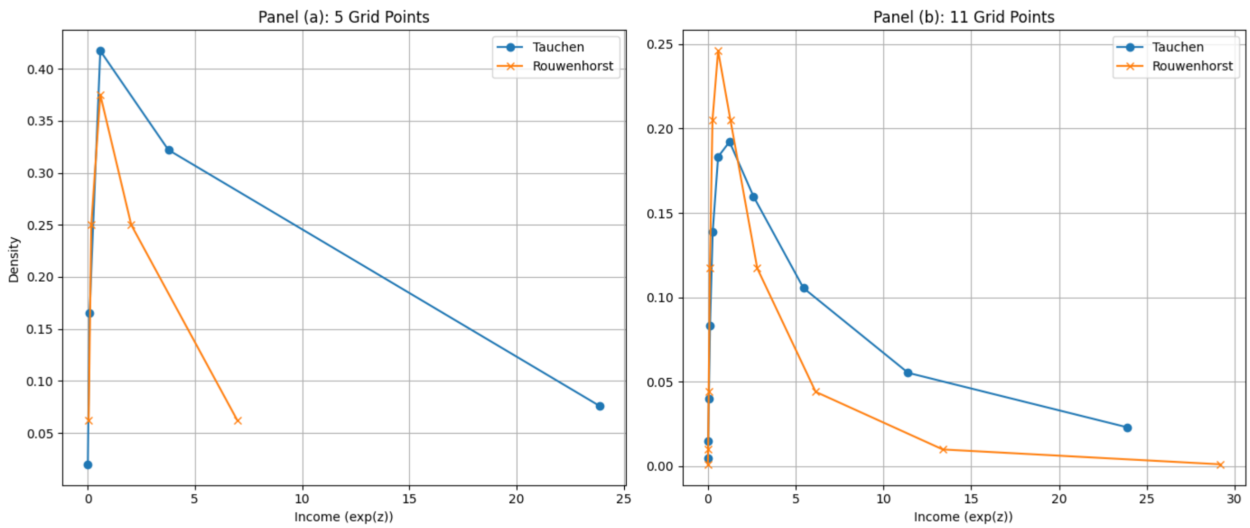


Note: In both simulations, I feed the same path of shocks by using a common seed

At $\rho = 0.9$, it is hard to distinguish if there is a clear approximation method that is better. All we can say is that indeed the approximation of the income dynamics and the stationary distribution of income (which is lognormal) are better

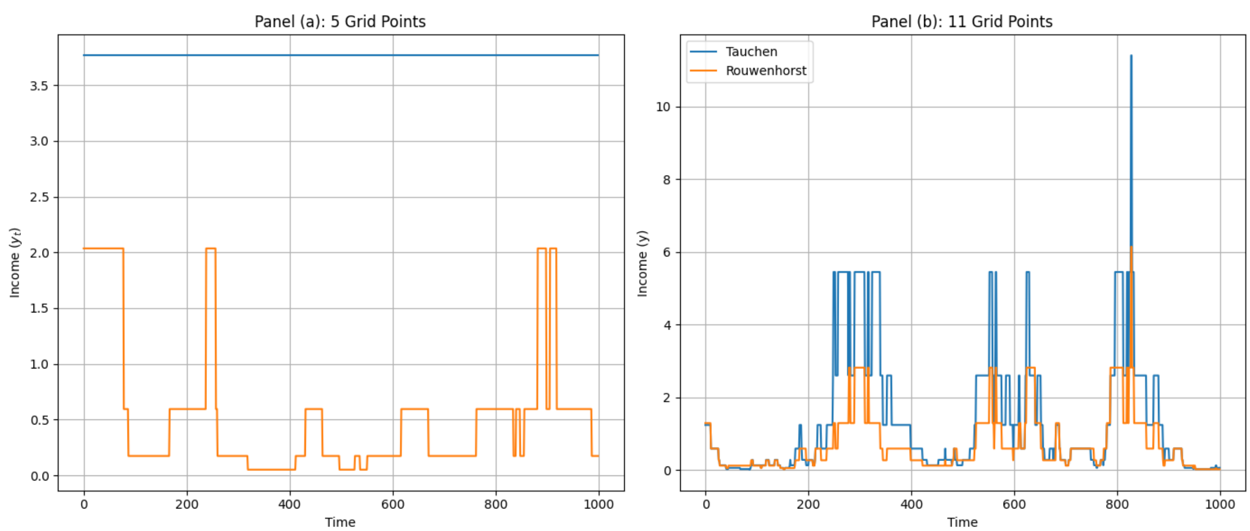
1.3 & 1.4 - Tauchen & Rouwenhorst

Figure 3: Stationary Distribution: Rouwhenhorst $\times$ Tauchen - $\rho = 0.98$



Note:I Compute the stationary distribution using the eigenvalue method, $\pi'(I - P) = 0$

Figure 4: Income Simulation: Rouwhenhorst $\times$ Tauchen - $\rho = 0.98$



Note: In both simulations, I feed the same path of shocks by using a common seed

At $\rho = 0.98$, the Tauchen method breaks down due to the high persistence for lower quantities of grid-points, performing poorly in simulation, as shown in Figure (4). However, for a bigger number of gridpoints, it still has a good performance even though below rouwenhorst

Question 2 - Income Fluctuation Problem

In this question, I analyze the numerical solution to the Income Fluctuation Problem. We have the usual framework, where we work with an income process with the Markov Property, implying we have a Markov Decision Problem. Thus, we have the following Recursive Formulation for our MDP:

$$V(a,s) = \max_{a' \geq \underline{b}} \{u((1+r)a + ws - a') + \beta \mathbb{E}[V(a',s')|s]\} \tag{1}$$

which yields an Euler Equation:

$$u'(c(a,s)) \geq \beta(1+r)\mathbb{E}[u'(c(a',s')) \mid s]$$

where the inequality holds when $a' > \underline{a}$ and not otherwise. We are interested in the consumption and next-period assets policy function which arises from the above problem, $c = c(a,s); a' = s(a,s)$. From the next-period assets, we can find the savings policy function $\text{sav}(a,s) = a'(a,s) - a$. Let $\mathcal{A}$ be the asset grid (which we build using the Power Grid method):

Definition (Power-Spaced Grid)**.** *A Power-spaced grid is a parametrized form of constructing a nonlinear grid $\mathcal{A}$. The grid may be nonlinear around its edges or a specific point of interest. For the first case, let $j \in \{1,\dots,N\}$ be the grid point of $\mathcal{A}$, the grid is built into three steps:*

(i) Construct a N-point equidistant grid $\mathcal{X}$ on the $[0,1]$ interval, where $x_1 = 0$ and $x_N = 1$

(ii) Make a non-linear transformation of the $\mathcal{X}$ grid to $\widetilde{\mathcal{X}}$ grid through a parameterized mapping, where ϕ controls the density of grid points on the lower part of the grid, whereas ν controls the density of the grid in the upper part of the grid:

$$\widetilde{x}_j = x_1 + \phi x_j^{\nu} \tag{2}$$

(iii) Apply a linear transformation to $\widetilde{\mathcal{X}}$, obtaining the desired $\mathcal{A}$ grid by using its extremities:

$$a_j = \underline{a} + \widetilde{x}_j \left(\frac{\bar{a} - \underline{a}}{\bar{x} - \underline{x}}\right) \tag{3}$$

and $\mathcal{S}$ be the income grid (which is built using the Rouwenhorst method). We discretize the above Value Function and Euler equation at the grid $\mathcal{A} \times \mathcal{S}$ as:

$$V(\bar{a}_i, \bar{s}_h) = \max_{a' \in \mathcal{A}} \left\{ u((1+r)\bar{a}_i + w\bar{s}_h - a') + \beta \sum_{j=1}^{m} \mathbf{P}(\bar{s}_j \mid \bar{s}_h) V(a', \bar{s}_j) \right\} \tag{4}$$

$$u'(R\bar{a}_i + w\bar{s}_h - a') \geq \beta(1+r) \sum_{j=1}^{m} \mathbf{P}(\bar{s}_j \mid \bar{s}_h) u'(Ra' + w\bar{s}_j - a'') \tag{5}$$

which will yield the policy functions $\{c(a,s), a'(a,s)\}$ in the $\mathcal{A} \times \mathcal{S}$ grid. To solve for this problem, we assume the following parameters as calibration:

2.1 - Plotting the Policy Functions

We work with three algorithms for solving the income Fluctuation Problem: (I) Value Function Iteration, (II) Policy Function Iteration, and (III) Endogenous Grid Method. The advantage of the Endogenous Grid Method is that it is as quick as the Value Function Iteration while being as accurate as the Policy Function Iteration.

Table 1: Baseline Calibration of the Partial Equilibrium Income Fluctuation Process

Description	**Value**
Discount Factor	$\beta = 0.95$
Interest Rate	$r = 2\%$
Inverse Elasticity of Substitution / RRA	$\gamma = 2$
Wage	$w = 1$
Borrowing Limit	$\underline{b} = 0$

Value Function Iteration

Definition (Value function Iteration)**.** *The Following Algorithm solves the Income Fluctuation Problem Numerically by Value Function Interation:*

1. *Initialize the following objects*
   - *m vectors which represent the value function at each point of the asset grid given a point on the income grid $\bar{s}_j$:*
   $$\mathbf{v}_j(i) = v(\bar{a}_i, \bar{s}_j) \quad \textit{for } i = 1, ..., n; \quad j = 1, ..., m$$
   - *m Matrices $n \times n$ for current period assets $\bar{a}_i$, next period a_h and current period income $\bar{s}_j$:*
   $$\mathbf{R}_j(i, h) = u((1+r)\bar{a}_i + w\bar{s}_j - \bar{a}_h) \quad \textit{for } i = 1, ..., n; \quad h = 1, \ldots, N; \quad j = 1, ..., m$$
   - *Column vector of ones: $\mathbf{1}_n$*
   - *Bellman operator $T([\mathbf{v}_1, ..., \mathbf{v}_m]) = [T\mathbf{v}_1, ..., T\mathbf{v}_m]$ such that:*
   $$T\mathbf{v}_j = \max\left\{\mathbf{R}_j + \beta \sum_{l=1}^{m} \mathbf{P}_{jl} \mathbf{1} \mathbf{v}_l'\right\}$$
   where the Right-Hand side inside the max operator is an $n \times n$ matrix, where each row represents the current level of assets and each column the next-period level of assets being chosen. The max *operator acts row-by-row on the matrix, returning a $n \times 1$ vector where the i^{th} element is the maximum of the i^{th} row. We can write the above equation in matrix form:*
   $$\mathbf{TV} = \max\left\{\begin{bmatrix}\mathbf{R}_1 \\ \mathbf{R}_2 \\ \vdots \\ \mathbf{R}_m\end{bmatrix} + \beta(\mathcal{P} \otimes \mathbf{1})\begin{bmatrix}\mathbf{v}_1' \\ \mathbf{v}_2' \\ \vdots \\ \mathbf{v}_m'\end{bmatrix}\right\} \tag{6}$$
   where $\mathbf{TV}$ is a $nm \times 1$ vector. Policy functions can be computed as follows:
   $$a(\bar{a}_i, \bar{s}_j) = \underset{a' \in \mathcal{A}}{\arg\max}\left\{\mathbf{R}_j + \beta \sum_{l=1}^{m} \mathbf{P}_{jl} \mathbf{1} \mathbf{v}_l'\right\} \tag{7}$$
   $$c(\bar{a}_i, \bar{s}_j) = w\bar{s}_j + R\bar{a}_i - a(\bar{a}_i, \bar{s}_j) \tag{8}$$
2. *Guess initial $V_0(a, s)$. Good guess is the value of consuming the permanent income,*
   $$V_0(a, s) = \sum_{t=0}^{\infty} \beta^t u(Ra + ws) = \frac{u(Ra + ws)}{1 - \beta}$$

3. *Using the Bellman Operator, compute the next iteration of the value function:*

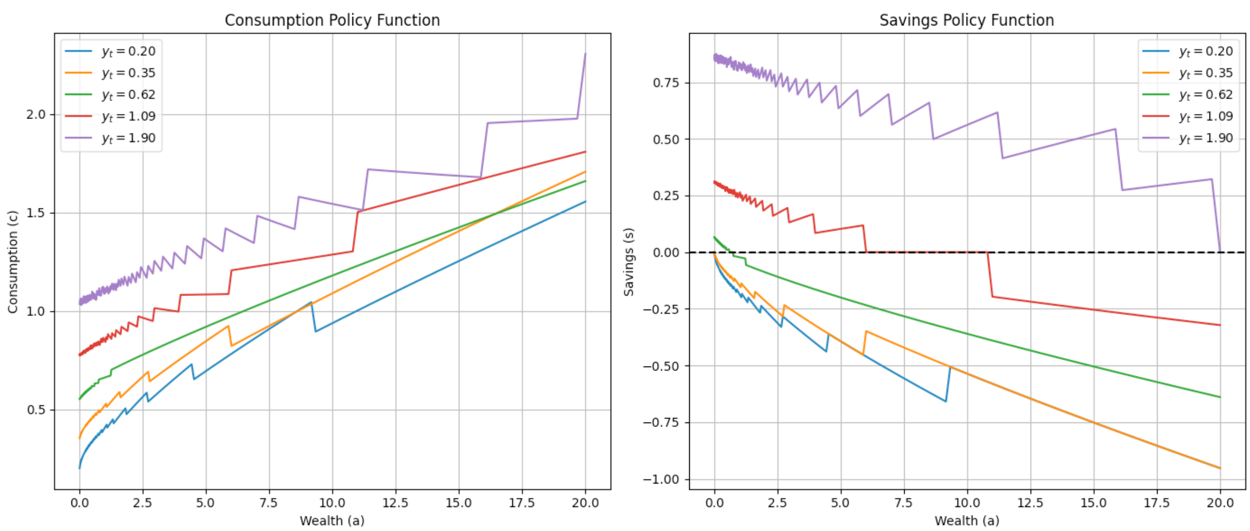
$$\mathbf{V}_{l+1} = \mathbf{T}\mathbf{V}_l \tag{9}$$

4. *Given a tolerance level ε (e.g. $\varepsilon = 10^{-5}$), check for convergence:*

$$\varepsilon_l = \max |\mathbf{V}_{l+1} - \mathbf{V}_l| \tag{10}$$

   if $\varepsilon_l > \varepsilon$, then return to the previous step. Otherwise, stop iteration as it has numerically converged

Figure 5: Solution of the IFP: Value Function Iteration



Note: Solution of the Income Fluctuation problem under the Value Function Iteration Algorithm ()

Policy Function Iteration

Definition (Policy Function Iteration)**.** *The following algorithm solves the Income Fluctuation Problem numerically by Policy Function Iteration using discretization and/or interpolation methods to solve for the next period assets. Take as given the state-space grids $\mathcal{A}$ and $\mathcal{S}$*

1. *Conjecture an initial decision rule $n \times m$ vector for a'':*

$$a_0''(\bar{a}_h, \bar{s}_i)$$

   one possible initial example is to set $a_0''(\bar{a}_h, \bar{s}_i) = \bar{a}_h$

2. *For each point $(\bar{a}_h, \bar{s}_i) \in \mathcal{A} \times \mathcal{S}$ in the grid, we check whether - conditional on $a_\ell''(\bar{a}_h, \bar{s}_i)$ - the borrowing constraint $\bar{a}_1 = \underline{a}$ is active. If the inequality is strict:*

$$u'(R\bar{a}_h + w\bar{s}_i - \underline{a}) > \beta R \sum_{j=1}^{m} \mathbf{P}(\bar{s}_j \mid \bar{s}_i) u'(R\underline{a} + w\bar{s}_j - a_\ell''(\underline{a}, \bar{s}_j)) \tag{11}$$

   then we set $a_\ell'(\bar{a}_h, \bar{s}_i) = \bar{a}_1$. Otherwise, if the above equation holds with a $<$ inequality, then there is an interior solution (it is optimal to save or borrow less than the borrowing limit) and we move to the next step

3. *Determine the interior solution of $a'_\ell(\bar{a}_h, \bar{s}_i)$. The main issue here is that the solution a^* not necessarily lies in the grid $\mathcal{A}$. Therefore, how can we evaluate $a''_\ell(\bar{a}_h, \bar{s}_i)$? We turn to each of the three variants:*

   - ***Discretization**: We look for the pair of adjacent grid points $(\bar{a}_k, \bar{a}_{k+1})$ such that:*

$$\delta(\bar{a}_k) \equiv u'(R\bar{a}_h + w\bar{s}_i - \bar{a}_k) - \beta R \sum_{j=1}^{m} \mathbf{P}(\bar{s}_j \mid \bar{s}_i)\, u'(R\bar{a}_k + w\bar{s}_j - a''_\ell(\bar{a}_k, \bar{s}_j)) < 0 \tag{12}$$

$$\delta(\bar{a}_{k+1}) \equiv u'(R\bar{a}_h + w\bar{s}_i - \bar{a}_{k+1}) - \beta R \sum_{j=1}^{m} \mathbf{P}(\bar{s}_j \mid \bar{s}_i)\, u'(R\bar{a}_{k+1} + w\bar{s}_j - a''_\ell(\bar{a}_{k+1}, \bar{s}_j)) > 0, \tag{13}$$

   and then we pick the one which is closest to zero:

$$a'_\ell(\bar{a}_h, \bar{s}_j) = \underset{i \in \{k, k+1\}}{\arg\min}\, |\delta(\bar{a}_i)| \tag{14}$$

   - ***Linear Interpolation**: For each point on the grid $(\bar{a}_h, \bar{s}_i)$, we use a nonlinear equation solver (e.g. Newton Method, Bissection, etc) to look for the solution a^* to the Euler Equation:*

$$u'(R\bar{a}_h + w\bar{s}_i - a^*) = \beta R \sum_{j=1}^{m} \mathbf{P}(\bar{s}_j \mid \bar{s}_i) u'(Ra^* + w\bar{s}_j - a''_\ell(a^*, \bar{s}_j))$$

   Say that a^* lies outside of $\mathcal{A}$. How can we compute $a''_\ell(a^*, \bar{s}_j)$? We approximate linearly the policy function $a''_\ell(a^*, \bar{s}_j)$ by the points $\bar{a}_k < a^* < \bar{a}_{k+1}$:

$$a''_\ell(a^*, \bar{s}_j) = a''_\ell(\bar{a}_k, \bar{s}_j) + (a^* - \bar{a}_k)\left(\frac{a''_\ell(\bar{a}_{k+1}, \bar{s}_j) - a''_\ell(\bar{a}_k, \bar{s}_j)}{\bar{a}_{k+1} - \bar{a}_k}\right) \tag{15}$$

   as we solve for a^*, the we use it to set the policy function:

$$a'_\ell(\bar{a}_h, \bar{s}_j) = a^* \tag{16}$$

4. *Check for convergence of the policy function given a tolerance parameter ε:*

$$\varepsilon_\ell = \max |\mathbf{a}'_l - \mathbf{a}''_l| \tag{17}$$

   if $\varepsilon_\ell \leq \varepsilon$, then stop the loop. Otherwise, set $a''_{\ell+1}(\bar{a}_h, \bar{s}_j) = a'_\ell(\bar{a}_h, \bar{s}_j)$

Endogenous Grid Method

Definition (Endogenous Grid Method)**.** *The following algorithm solves the Income Fluctuation Problem numerically by the Endogenous Grid Method using interpolation methods to solve for current-period consumption. Take as given the state-space grids $\mathcal{A}$ and $\mathcal{S}$:*

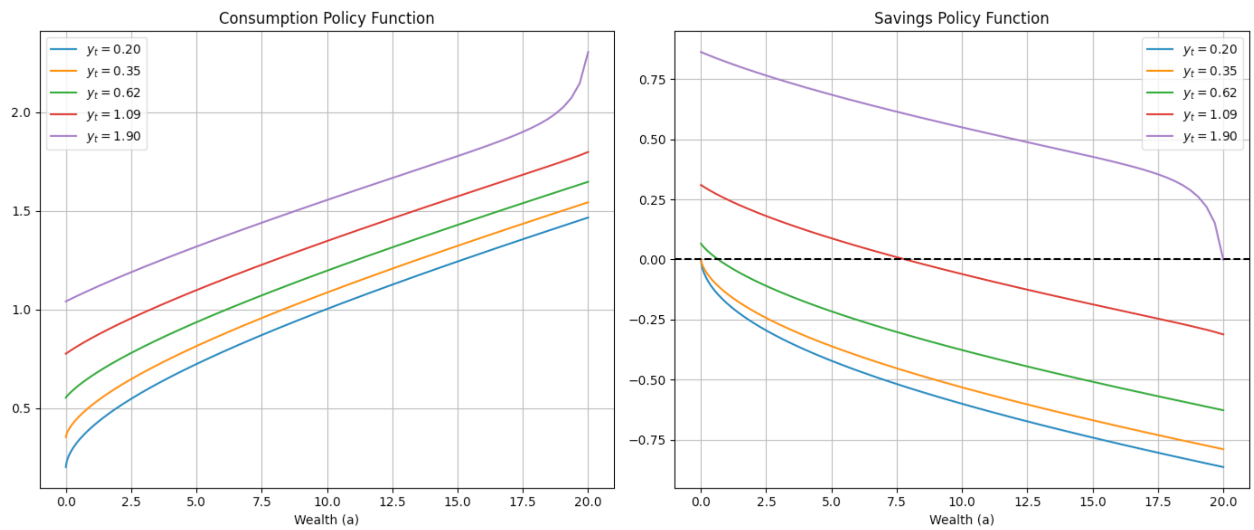
1. *Conjecture an initial decision rule $n \times m$ for current-period consumption $c_0(\bar{a}_h, \bar{s}_i)$. One example is:*

$$c_0(\bar{a}_h, \bar{s}_i) = R\bar{a}_h + w\bar{s}_i$$

2. *Given the current period consumption function $c_\ell(\bar{a}_h, \bar{s}_i)$, we want to find the endogenous grid of assets $a^*(\bar{a}'_h, \bar{s}_i)$ - that is, a function of tomorrow assets $\bar{a}'_h \in \mathcal{A}$ and today's income $\bar{s}_i \in \mathcal{S}$ which delivers the level of current assets a^* that would lead to $\bar{a}'_h$ if my income today is $\bar{s}_i$. To obtain it, first we find today's consumption as a function of tomorrow's assets and today's income through the Euler equation:*

$$c^{FOC}(\bar{a}'_h, \bar{s}_i) = (u')^{-1}\left(\beta R \sum_{j=1} \mathbf{P}(\bar{s}_j \mid \bar{s}_i) u'(c_\ell(\bar{a}'_h, \bar{s}_j))\right) \tag{18}$$

Figure 6: Solution of the IFP: Policy Function Iteration



Note: Solution of the Income Fluctuation problem under the Policy Function Iteration Algorithm ()

Then, from the budget constraint, obtain $a^*(\bar{a}'_h, \bar{s}_i)$*:*

$$a^*(\bar{a}'_h, \bar{s}_i) = \frac{1}{R}\left(c_\ell^{FOC}(\bar{a}'_h, \bar{s}_i) + \bar{a}'_h - w\bar{s}_i\right) \tag{19}$$

Notice that, implicitly, we are obtaining the next step function $c_{\ell+1}(a^*(\bar{a}'_h, \bar{s}_i), \bar{s}_i) = c^{FOC}(\bar{a}'_h, \bar{s}_i)$*. However, this function is not defined on the grid* $\mathcal{A}$*, rather on the endogenous grid* $\{a^*(\bar{a}'_h, \bar{s}_i)\}_{h,i}$*. The last step is to map this function to the original grid* $\mathcal{A} \times \mathcal{S}$ *and take into account the borrowing constraint, which we ignored when calculating* c^{FOC}

3. *Given the grid* $\mathcal{A} \times \mathcal{S}$ *and the function* a^**, we do the following loop to find* $c_{\ell+1}(\bar{a}_h, \bar{s}_i)$ *conditional on a fixed the income* $\bar{s}_i$*:*

   3.1 *Find the asset level today which implies that the agent is constrained tomorrow,* $a_1^* \equiv a^*(\underline{a}, \bar{s}_i)$

   3.2 *For all* $\bar{a}_h \in \mathcal{A}$ *such that* $\bar{a}_h \leq a_1^*$*, set the consumption of the household as hand-to-mouth:*

$$\begin{aligned} c_{\ell+1}(\bar{a}_h, \bar{s}_i) &= R\bar{a}_h + w\bar{s}_i - \underline{a}, \quad \forall\ \bar{a}_h \leq a_1^* \\ a'_{\ell+1}(\bar{a}_h, \bar{s}_i) &= \underline{a} \end{aligned}$$

   3.3 *For all* $\bar{a}_h \in \mathcal{A}$ *such that* $\bar{a}_h > a_1^*$*, find the* $c_{\ell+1}(\bar{a}_h, \bar{s}_i)$ *through interpolation: Find* $\{a_k^*, a_{k+1}^*\}$ *on the endogenous grid such that* $a_k^* < \bar{a}_h < a_{k+1}^*$*, and interpolate using* $\{c^{FOC}(a_k^*, \bar{s}_i); c^{FOC}(a_{k+1}^*, \bar{s}_i)\}$*:*

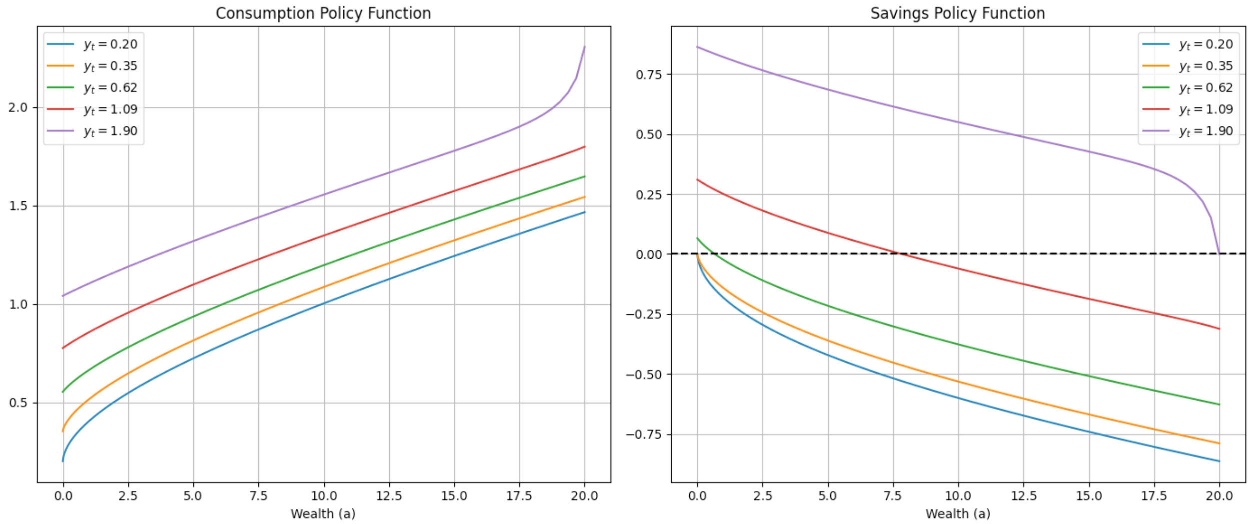
$$\begin{aligned} c_{\ell+1}(\bar{a}_h, \bar{s}_i) &= c^{FOC}(a_k^*, \bar{s}_i) + (\bar{a}_h - a_k^*)\left(\frac{c^{FOC}(a_{k+1}^*, \bar{s}_i) - c^{FOC}(a_k^*, \bar{s}_i)}{a_{k+1}^* - a_k^*}\right) \\ a_{\ell+1}(\bar{a}_h, \bar{s}_i) &= R\bar{a}_h + w\bar{s}_i - c_{\ell+1}(\bar{a}_h, \bar{s}_i) \end{aligned}$$

4. *Check for convergence of the policy function given a tolerance parameter* ε*:*

$$\varepsilon_\ell = \max|\mathbf{c}_{\ell+1} - \mathbf{c}_\ell| \tag{20}$$

   if $\varepsilon_\ell \leq \varepsilon$*, then stop the loop. Otherwise, go back to step 2 with the new policy function*

Figure 7: Solution of the IFP: Endogenous Grid Method



Note: Solution of the Income Fluctuation problem under the Endogenous Grid Method Algorithm ()

2.2 - Simulation and Accuracy

The VFI method is the least accurate, in spite of the sizable quantity of points in the asset grid $n_a = 300$. This happens as the power grid creates sizable distortions as the assets of the household grow larger. These distortions are smaller for lower levels of assets. The PFI and the EGM methods are virtually indistinguishable. Paths for consumption, savings and assets of a simulated household are available in the figure (8)

2.3 - Consumption and Risk Aversion

As the relative risk aversion parameter γ increases, households increase their intertemporal smoothing and precautionary motive, where both act in the sense of decreasing the volatility in the individual consumption. This can be seen in the figure (9), where the volatility of consumption decreases with an increase in the risk aversion parameter.

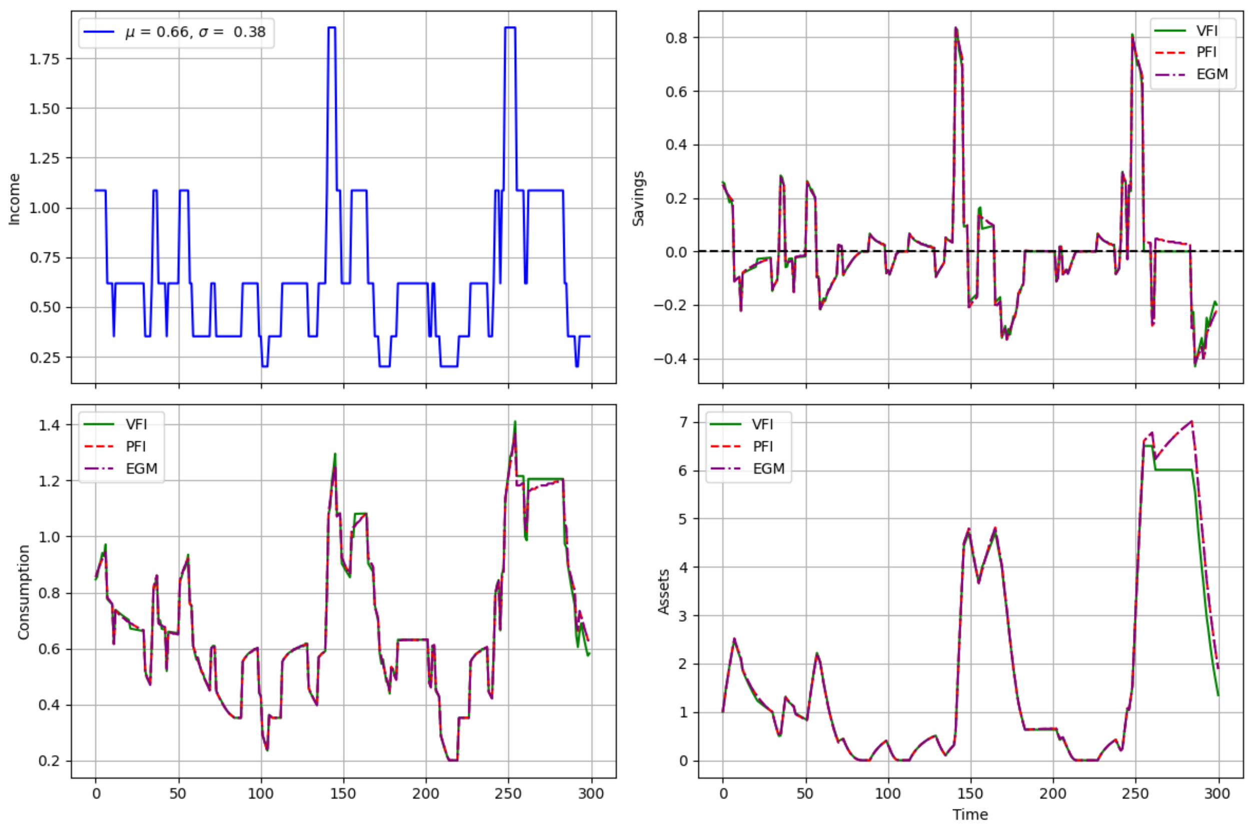
2.4 - Saving-Income Ratio

As we increase the variance of innovations to the income shock, the saving income ratio becomes more-volatile and decreases in mean (agents will dissave more in average). This can be seen in the figure (10)

2.5 - Natural Borrowing Limit

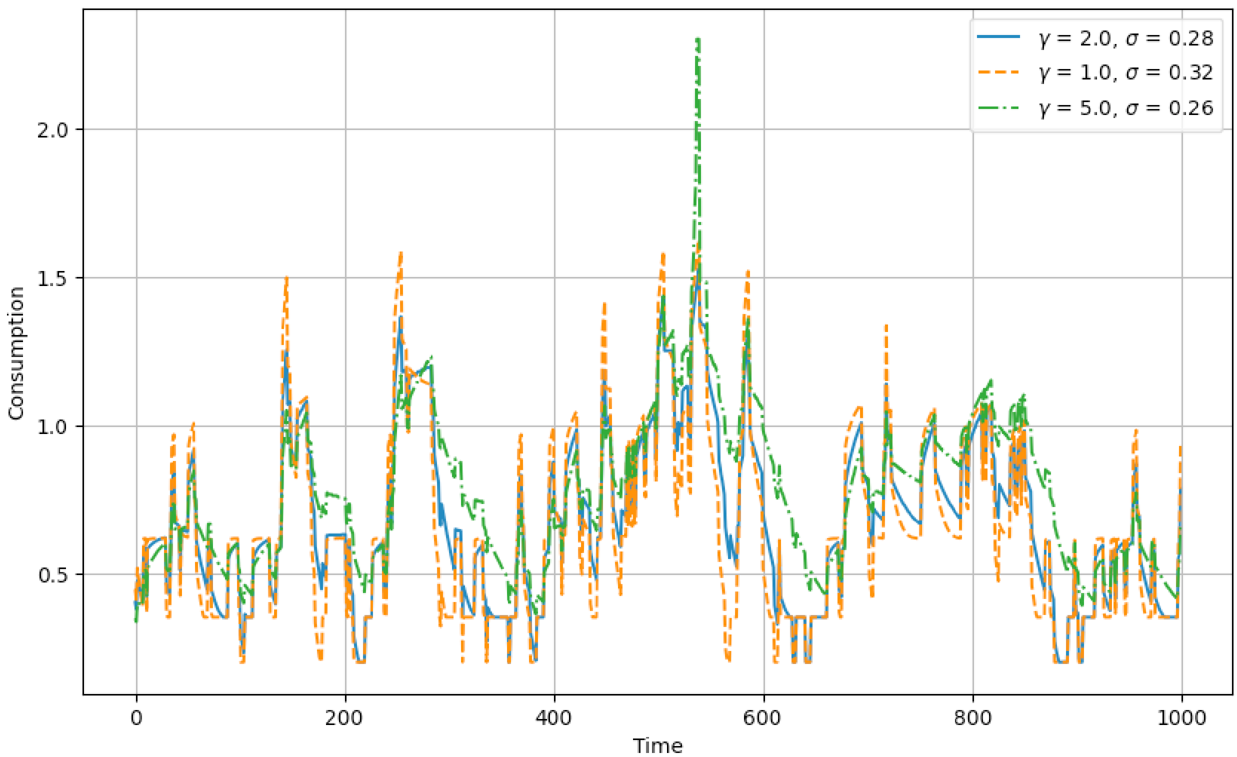
In the presence of a natural borrowing limit, consumption is higher and smoother, in the sense that as agents have more access to insurance so they consume more and have lower volatility of their consumption path. This can be seen in figure (11), as the consumption stochastic process has lower volatility and a higher mean - opposite to the savings process. The more marked difference lies on the asset path, where the impatience of the household is even more clear as it runs down assets into negative territory in the face of negative income shocks

Figure 8: Simulation of Household Consumption, Savings, Consumption and Assets under different Solution Methods



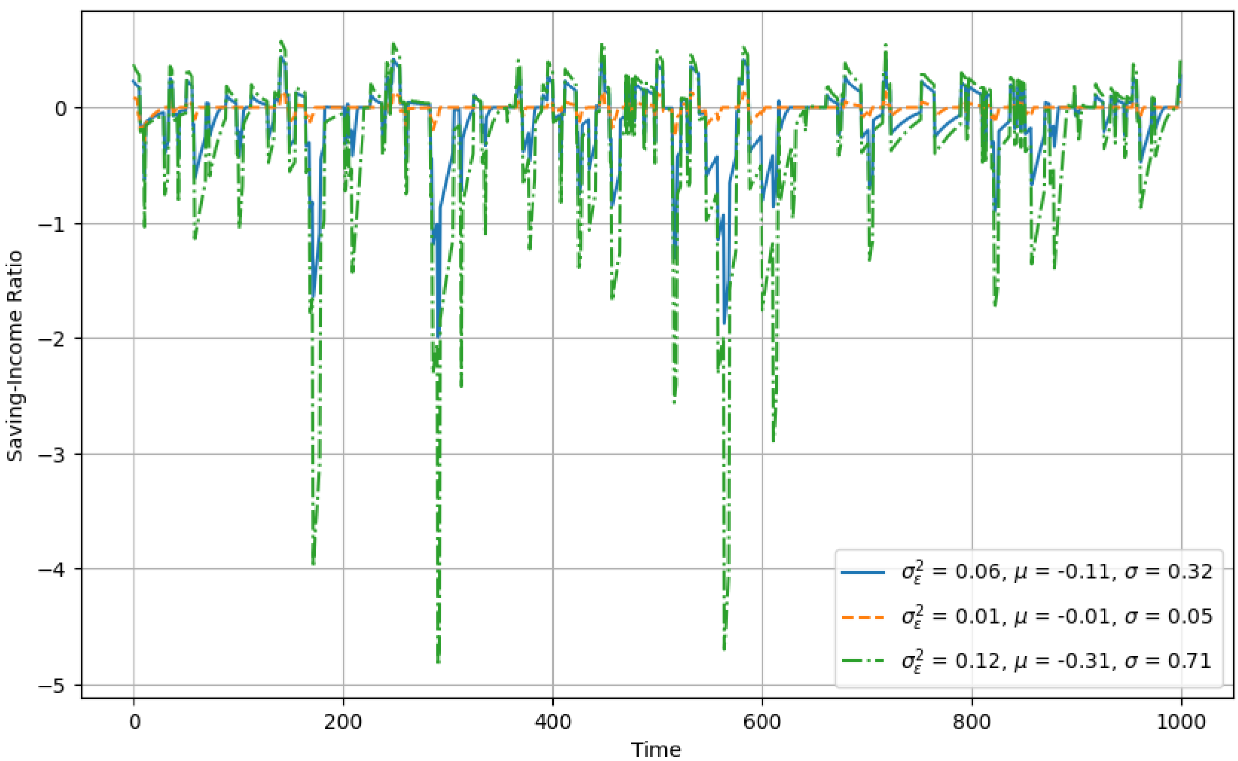
Note: We simulate the path of consumption for $t = 300$ periods under the solution of the Income Fluctuation Problem for each of the three methods (VFI, PFI, EGM) under the same calibration as in table (1) with the AR(1) income process calibration. We set the same seed in the code, `seed = 42`, to ensure the sequence of income y_t is the same across simulations

Figure 9: Simulation of Household Consumption Under Different Risk Aversion



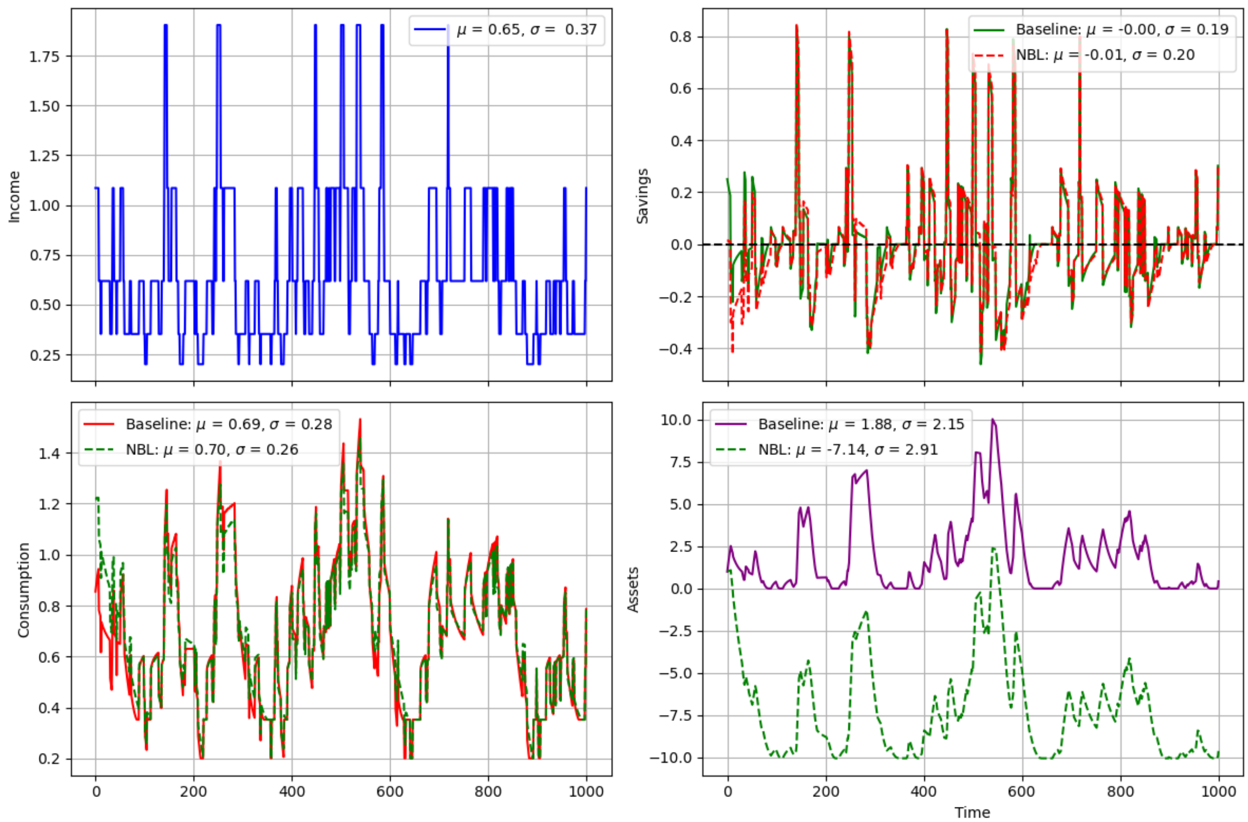
Note: We simulate the path of consumption for $t = 1000$ periods under the solution of the Income Fluctuation Problem under different levels of risk aversion $\gamma \in \{1, 2, 5\}$ with the Endogenous Grid Method and Calibration as in table (1) with the AR(1) income process calibration. We set the same seed in the code, `seed = 42`, to ensure the sequence of income y_t is the same across simulations

Figure 10: Simulation of Savings to Income Ratio Under different σ_ε^2



Note: We simulate the path of savings to income ratio for $t = 1000$ periods under the solution of the Income Fluctuation Problem under different levels of variance to the innovations to the income shock $\sigma_\varepsilon^2 \in \{0.01, 0.06, -.12\}$ with the Endogenous Grid Method and Calibration as in table (1) with the AR(1) income process calibration. We set the same seed in the code, `seed = 42`, to ensure the sequence of income y_t is the same across simulations

Figure 11: Simulation of Household Behavior under No-Borrowing Limit × Natural Borrowing Limit



Note: We simulate the path of Consumption, Savings and Assets for $t = 1000$ periods under the solution of the Income Fluctuation Problem under the no-borrowing limit $\underline{b} = 0$ and the natural borrowing limit $\underline{b} = -\frac{y_{min}}{r}$ with the Endogenous Grid Method and Calibration as in table (1) with the AR(1) income process calibration. We set the same seed in the code, `seed = 42`, to ensure the sequence of income y_t is the same across simulations